
Sample complexity of Schrödinger potential estimation

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Abstract

1 We address the problem of Schrödinger potential estimation, which plays a crucial
2 role in modern generative modelling approaches based on Schrödinger bridges and
3 stochastic optimal control for SDEs. Given a simple prior diffusion process, these
4 methods search for a path between two given distributions ρ_0 and ρ_T requiring min-
5 imal efforts. The optimal drift in this case can be expressed through a Schrödinger
6 potential. In the present paper, we study generalization ability of an empirical
7 Kullback-Leibler (KL) risk minimizer over a class of admissible log-potentials
8 aimed at fitting the marginal distribution at time T . Under reasonable assumptions
9 on the target distribution ρ_T and the prior process, we derive a non-asymptotic
10 high-probability upper bound on the KL-divergence between ρ_T and the terminal
11 density corresponding to the estimated log-potential. In particular, we show that
12 the excess KL-risk may decrease as fast as $\mathcal{O}(\log n/n)$ when the sample size n
13 tends to infinity even if both ρ_0 and ρ_T have unbounded supports.

14 1 Introduction

15 The Schrödinger Bridge problem (SBP) originates from a question posed by Erwin Schrödinger in
16 1932 [Schrödinger, 1932], seeking the most likely evolution of a probability distribution between
17 two given endpoint distributions while minimizing relative entropy with respect to a prior stochastic
18 process. This problem has deep connections with optimal transport [Leonard, 2014] and stochastic
19 control [Dai Pra, 1991]. In its simplest continuous-time form, one aims to construct a so-called
20 *Schrödinger Markov process* whose joint begin-end distribution $\pi(dx, dz)$ has the representation

$$\pi(dx, dz) = Q(z, T \mid x, 0) \nu_0(dx) \nu_T(dz), \quad (1)$$

21 where $Q(z, T \mid x, 0)$ is the transition kernel of a reference Markov process, and ν_0, ν_T are unknown
22 “boundary potentials” to be determined. The desired marginals $\pi(dx, \mathbb{R}^d)$ and $\pi(\mathbb{R}^d, dz)$ are given,
23 and one seeks ν_0 and ν_T that reproduce these marginals. In the rest of the paper, we assume that
24 both $\pi(dx, \mathbb{R}^d)$ and $\pi(\mathbb{R}^d, dz)$ are absolutely continuous with respect to the Lebesgue measure and
25 denote the corresponding densities by ρ_0 and ρ_T^* , respectively. Classical existence proofs for the SBP
26 date back to Fortet [1940] (in 1D) and Beurling [1960], with a modern fixed-point approach in [Chen
27 et al., 2016]. Recent extensions to the case of noncompactly supported marginal distributions can be
28 found in [Conforti et al., 2024] and [Eckstein, 2025]. Recently, the problem attracted attention of
29 machine learners in the context of generative modelling (see, for instance, [Tzen and Raginsky, 2019,
30 De Bortoli et al., 2021, Shi et al., 2023, Korotin et al., 2024, Gushchin et al., 2024a, Rapakoulis
31 et al., 2024] to name a few). It follows from Theorem 3.2 in [Dai Pra, 1991] that the optimal Markov
32 process X_t^* solving the Schrödinger problem with marginals (ρ_0, ρ_T^*) can be constructed as a solution
33 of the following SDE:

$$dX_t^* = (b(X_t^*, t) + \sigma(X_t^*, t)\sigma(X_t^*, t)^\top \nabla \log h(X_t^*, t)) dt + \sigma(X_t^*, t) dW_t, \quad X_0 \sim \rho_0,$$

34 where

$$h(w, t) = \int_{\mathbb{R}^d} Q(y, T \mid w, t) \nu_T(dy)$$

35 and Q is the transition density of the reference (or base) diffusion process

$$dX_t = b(X_t, t) dt + \sigma(X_t, t) dW_t, \quad X_0 \sim \rho_0.$$

36 The transition density Q^* of the reciprocal process X_t^* can be obtained from Q via the so-called
37 Doob's h -transform:

$$Q^*(y, T \mid x, t) = Q(y, T \mid x, t) \frac{h(y, T)}{h(x, t)}. \quad (2)$$

38 This is precisely the law of the base process conditioned by the function h (see [Jamison, 1974]). In
39 many presentations of the Schrödinger Bridge problem, one takes a very simple reference process (for
40 instance, a Brownian motion) so that its transition kernel is straightforward to write down (see, for
41 example, [Pooladian and Niles-Weed, 2024] and [Baptista et al., 2024]). However, there are several
42 practical and theoretical advantages to considering more general (potentially higher-dimensional, or
43 with domain constraints, or with a non-trivial drift/diffusion) reference processes.

44 In the present paper, we are interested in estimation of the Schrödinger potential ν_T from n i.i.d.
45 samples $Y_1, \dots, Y_n \sim \rho_T^*$. Given a class of log-potentials Ψ , we study generalization ability of an
46 empirical risk minimizer

$$\hat{\psi} \in \operatorname{argmin}_{\psi \in \Psi} \left\{ -\frac{1}{n} \sum_{i=1}^n \log \left(\int_{\mathbb{R}^d} Q(Y_i, T \mid x, 0) \frac{h_\psi(Y_i, T)}{h_\psi(x, 0)} \rho_0(x) dx \right) \right\}, \quad (3)$$

47 where

$$h_\psi(x, t) = \int Q(y, T \mid x, t) e^{\psi(y)} dy.$$

48 Let us note that, in view of (2),

$$\rho_T^\psi(y) = \int_{\mathbb{R}^d} Q(y, T \mid x, 0) \frac{h_\psi(y, T)}{h_\psi(x, 0)} \rho_0(x) dx$$

49 is the marginal endpoint probability density of a diffusion process X_t^ψ corresponding to Doob's
50 h_ψ -transform:

$$dX_t^\psi = \left(b(X_t^\psi, t) + \sigma(X_t^\psi, t) \sigma(X_t^\psi, t)^\top \nabla \log h_\psi(X_t^\psi, t) \right) dt + \sigma(X_t^\psi, t) dW_t, \quad X_0 \sim \rho_0.$$

51 In other words, the estimate $\hat{\psi}$ minimizes empirical Kullback-Leibler (KL) divergence between the
52 actual target ρ_T^* and the marginal densities ρ_T^ψ over the class of admissible log-potentials Ψ . That is,
53 we chose the log-potential ψ that makes the transformed reference diffusion hit the observed terminal
54 law, and measure error only through KL of the marginals. Because h_ψ is used inside the Doob factor,
55 the learnt potential is compatible with a single Markov process; one never risks obtaining mutually
56 inconsistent forward/backward potentials. The method combines the full problem (the marginals,
57 transition densities, and the potential function) into one single optimization framework. By doing so,
58 it aims to directly minimize the objective of matching the marginals at time T without separating the
59 problem into smaller subproblems. In contrast, the Sinkhorn algorithm, commonly used for optimal
60 transport problems, approaches the problem by iteratively updating the potentials in a decoupled
61 manner. At each iteration, a simpler least squares problem appears, which is linear in one potential
62 function given that another one is fixed from the previous iteration. The Sinkhorn algorithm alternates
63 between updating the potential functions to match the marginals of the distributions and adjusting the
64 transport plan until convergence. We refer to [Pooladian and Niles-Weed, 2024], [Chiarini et al., 2024]
65 for recent results. The primary advantage of the Sinkhorn approach is its computational efficiency.
66 By decoupling the optimization process into simpler, linear problems, the Sinkhorn method can
67 handle large-scale problems effectively. This iterative procedure allows for faster updates, and it has
68 become a popular method for many optimal transport applications, see [Genevay et al., 2018], [March
69 and Henry-Labordere, 2023]. However, the approach presented in this paper differs in that it does
70 not separate the problem into independent steps. Instead, it aims at solving the Schrödinger system

approximately by formulating it as a single optimization problem involving Doob h -transform of the base process X parametrized by the Schrödinger potential. Unlike iterative proportional fitting (Sinkhorn), everything is learnt in one go, avoiding slow or unstable fixed-point cycles. This results in a more accurate and robust solution. The trade-off between the two methods lies in computational efficiency versus the quality of the solution. The Sinkhorn approach provides a quick and efficient solution by solving simpler problems at each iteration, but it may not achieve the best possible solution for the full problem. On the other hand, the method presented in this paper offers a more holistic approach, which could lead to a more accurate matching of the marginal distributions but might require more computational resources.

The approach presented in this paper can also be compared to methods that rely on optimization over transport maps, see [Korotin et al. \[2024\]](#), [Gushchin et al. \[2024a\]](#). In transport map-based approaches, the goal is to find a map $\mathcal{T}$ that transports one probability distribution to another. The optimization typically focuses on minimizing a quadratic cost functional that penalizes the difference between the target distribution and the transformed distribution under the transport map. These methods are often framed as optimal transport problems, where the map $\mathcal{T}$ is determined by solving an optimization problem that involves the marginal distributions. The advantage of optimization over transport maps lies in its clear geometric interpretation, where the transport map provides a direct way to relate the two distributions. This can lead to efficient algorithms, especially when the transport map can be parametrized in a way that allows for fast computations, such as in the case of certain neural network architectures or simple affine transformations, [Rapakoulis et al. \[2024\]](#).

However, transport map-based approaches are typically constrained to quadratic costs, which may limit their applicability in some cases. Specifically, quadratic cost functionals, such as the 2-Wasserstein distance, often assume a certain structure or symmetry that may not be ideal for more general or complex problems.

In contrast, the approach discussed in this paper is not limited to quadratic costs. It allows for more general cost structures and is based on minimizing the Kullback-Leibler divergence (KL-divergence), which can accommodate a wider range of problem types. This flexibility is particularly valuable when dealing with more complex distributions or when the underlying problem involves non-quadratic costs that capture other aspects of the distribution, such as entropy regularization or non-linear interactions between variables.

Contribution The main contribution of the present paper is a sharper non-asymptotic high-probability upper bound on generalization error of the empirical risk minimizer $\hat{\psi}$ defined in [\(3\)](#).

- Taking a multivariate Ornstein-Uhlenbeck process as the reference one, we show that (see [Theorem 1](#)), with probability at least $(1 - 2\delta)$, the excess KL-risk of the marginal endpoint density $\hat{\rho}_T$ corresponding to $\hat{\psi}$ satisfies the inequality

$$\text{KL}(\rho_T^*, \hat{\rho}_T) - \inf_{\psi \in \Psi} \text{KL}(\rho_T^*, \rho_T^\psi) \lesssim \sqrt{\Upsilon(n, \delta) \inf_{\psi \in \Psi} \text{KL}(\rho_T^*, \rho_T^\psi)} + \Upsilon(n, \delta),$$

where

$$\Upsilon(n, \delta) \lesssim \frac{\log^2 n + \log(1/\delta) \log n}{n}.$$

Here and further in the paper, the sign $\lesssim$ stands for an inequality up to a multiplicative constant. The derived upper bound has several advantages over the existing results. First, in contrast to [Korotin et al. \[2024\]](#), the excess risk may decrease as fast as $\mathcal{O}(\log^2 n/n)$ provided that the class of log-potentials Ψ is rich enough to approximate the target density ρ_T^* . Second, unlike theoretical guarantees for Sinkhorn-based approaches (see e.g. [Pooladian and Niles-Weed \[2024\]](#)), we are able to relate the endpoint marginal densities ρ_T^* and $\hat{\rho}_T$.

- We impose very mild assumptions on the target density ρ_T^* . We only require ρ_T^* to be bounded and sub-Gaussian. On the other hand, the available convergence proofs for the Sinkhorn algorithm rely on the stronger assumption that the marginals are log-concave, see [Conforti et al. \[2024\]](#). We also avoid the so-called strong density assumptions like boundedness from below often used in nonparametric statistics in the context of log-density estimation.
- The assumptions on the class of log-potentials Ψ are also reasonable. We support our claim with several examples.

Paper structure The rest of the paper is organized as follows. Section 2 is devoted to a short review of related work. In Section 3, we introduce necessary definitions and notations. After that, we present our main result (Theorem 1) in Section 4 and discuss main ideas of its proof in Section 5. Rigorous derivations as well as auxiliary technical results are deferred to the supplementary material.

2 Related work

Here is a short review of methods used in the literature to compute Schrödinger potentials, including the Sinkhorn algorithm. The Schrödinger potential, which arises in optimal transport problems, represents a key component in the solution of transport problems involving marginal distributions. Over time, several methods have been proposed to compute these potentials efficiently, with applications in areas ranging from statistical mechanics to machine learning. Here, we review some of the most prominent methods used in the literature.

Sinkhorn algorithm The Sinkhorn algorithm [Sinkhorn (1967)] is one of the most widely used methods for computing Schrödinger potentials in the context of optimal transport. It is based on iterative scaling and aims to solve the optimal transport problem by alternating between updating two potentials ν_0 and ν_T to enforce marginal constraints. The key advantage of the Sinkhorn approach is its computational efficiency, particularly when the transport problem is framed with a quadratic cost (such as the 2-Wasserstein distance), see [Pavon et al. (2021), Chen et al. (2021), Stromme (2023)] for reference. In each iteration, the algorithm solves a simpler problem that involves scaling the potentials in a way that brings the marginals of the transformed distribution closer to the target. Although Sinkhorn’s algorithm is efficient and widely applicable, it is often limited by its assumption of quadratic costs. Additionally, the algorithm does not directly handle more complex cost structures, such as non-quadratic costs or non-linear dynamics, which can be a limitation in some applications.

Sinkhorn bridge The Sinkhorn Bridge proposed by [Pooladian and Niles-Weed (2024)], provides a way to estimate the Schrödinger bridge using Sinkhorn’s algorithm in an efficient manner. The key insight of this method is that the potentials obtained from the static entropic optimal transport problem can be modified to yield a natural plug-in estimator for the drift function that defines the Schrödinger bridge. However, this work does not provide bounds on the distance between marginal distributions at time $T = 1$ because there is an exploding term $(1 - \tau)^{k+2}$ as $\tau \rightarrow 1$ where k is the dimension of the underlying manifold. This term leads to a “curse of dimensionality” where the error grows rapidly as τ approaches 1, especially in high-dimensional settings. As a result, the estimation error increases significantly when attempting to estimate the Schrödinger bridge at the terminal time, making it difficult to obtain precise bounds for $T = 1$.

Dual Formulation of the Schrödinger Problem In the dual formulation of the Schrödinger problem, the Schrödinger potential is computed by solving a convex optimization problem. This approach reformulates the problem in terms of a dual objective that involves the Kullback-Leibler (KL) divergence between the target and predicted distributions. The dual problem is then solved using optimization techniques such as gradient descent or variational methods, see [Zhang and Chen (2022), Tzen and Raginsky (2019)] for reference. This formulation is more flexible than the Sinkhorn algorithm, as it can accommodate more general cost functions and is not limited to quadratic losses.

While the dual approach is flexible, it is often computationally more demanding than Sinkhorn’s method due to the need for iterative optimization over high-dimensional spaces. This makes the dual formulation suitable for smaller or more specialized problems, but it can become computationally expensive in large-scale applications.

Approximate Solutions Using Monte Carlo Methods Monte Carlo methods, particularly those relying on reverse diffusion processes, have also been employed to approximate Schrödinger potentials. In these methods, a reverse process is simulated, and the potential is iteratively refined to minimize the discrepancy between the predicted and target marginals, see [Korotin et al. (2024)] for reference. These methods are often used when the problem involves complex dynamics that are difficult to capture using direct optimization techniques.

170 Monte Carlo methods are particularly useful when dealing with high-dimensional problems, as they
 171 allow for the sampling of large spaces. However, they can be computationally expensive and may
 172 require a significant number of samples to achieve an accurate solution.

173 In addition, there are approaches that rely heavily on Monte Carlo approximations of intermediate
 174 values rather than the Schrödinger potentials themselves, among which the following should be noted
 175 [De Bortoli et al. \[2021\]](#), [Vargas et al. \[2021\]](#), [Peluchetti \[2023\]](#).

176 **Neural Network-Based Approaches** Recent advancements in deep learning have led to the use of
 177 neural networks to approximate Schrödinger potentials. These approaches treat the potential function
 178 as a parameterized neural network and use gradient-based optimization techniques to learn the
 179 potential that best matches the marginals. The use of neural networks offers a flexible and powerful
 180 way to model complex non-linear potentials, making these methods well-suited for problems with
 181 intricate dynamics or non-quadratic costs. While neural network-based approaches are highly flexible,
 182 they require large amounts of data and computational resources to train the network, and they are
 183 often prone to overfitting if not regularized appropriately. Despite these challenges, they represent a
 184 promising direction for future research, especially when the problem at hand involves complex and
 185 high-dimensional systems. We refer to [Liu et al. \[2023\]](#), [Wang et al. \[2021\]](#) for recent results.

186 **Iterative Markovian Fitting** The Iterative Markovian Fitting (IMF) method, introduced in the
 187 recent work by [Shi et al. \[2023\]](#), offers an approach to solving Schrödinger Bridge (SB) problems.
 188 Unlike previous methods, such as Iterative Proportional Fitting (IPF), IMF guarantees the preservation
 189 of both the initial and terminal distributions in each iteration, which is a key advantage over IPF
 190 where these marginals are not always preserved. IMF alternates between two types of projections:
 191 Markovian projections and reciprocal projections, ensuring that the resulting distribution remains
 192 within the correct class (Markovian or reciprocal) while progressively approximating the Schrödinger
 193 Bridge. We refer to [Gushchin et al. \[2024b\]](#) for recent results.

194 In [Silveri et al. \[2024\]](#), the authors provide the convergence analysis for diffusion flow matching
 195 (DFM), a method used to generate approximate samples from a target distribution by bridging it with a
 196 base distribution through diffusion dynamics. Their theoretical work includes non-asymptotic bounds
 197 on the Kullback-Leibler (KL) divergence between the true target distribution and the distribution
 198 generated by the DFM model. A key insight from this paper is the incorporation of two sources
 199 of error: drift-estimation and time-discretization errors. However, while the convergence analysis
 200 offers theoretical guarantees, the statistical error is not explicitly addressed in this paper. The analysis
 201 assumes that all expectations are exact, which might not hold in practical settings where samples are
 202 finite, and statistical errors could arise due to the approximations involved in the generative process.
 203 Thus, future work will need to extend this analysis to quantify the impact of statistical approximations
 204 in finite-sample settings.

205 3 Preliminaries and notations

206 This section collects necessary definitions and notations. As we announced in the contribution
 207 paragraph, we are going to consider a multivariate Ornstein-Uhlenbeck process as a reference one.
 208 For this reason, we elaborate on its basic properties in this section.

209 **Multivariate Ornstein-Uhlenbeck process** To be more specific, we will consider the base process
 210 X_t^0 solving the SDE

$$dX_t^0 = b(m - X_t^0) dt + \Sigma^{1/2} dW_t, \quad 0 \leq t \leq T,$$

211 where $b > 0$ controls the drift rate, $m \in \mathbb{R}^d$ represents the mean-reversion level, $\Sigma \in \mathbb{R}^{d \times d}$ is a
 212 positive definite symmetric matrix, and W_t is a standard d -dimensional Wiener process. It is known
 213 that the conditional distribution of X_t^0 given $X_0^0 = x$ is Gaussian $\mathcal{N}(m_t(x), \Sigma_t)$ with

$$m_t(x) = (1 - e^{-bt})m + e^{-bt}x \quad \text{and} \quad \Sigma_t = \frac{1 - e^{-2bt}}{2b} \Sigma. \quad (4)$$

214 This implies that the corresponding Doob's h -transform can be expressed through the Ornstein-
 215 Uhlenbeck operator

$$\mathcal{T}_t g(x) = \frac{1}{(2\pi)^{d/2} \sqrt{\det(\Sigma_t)}} \int_{\mathbb{R}^d} \exp \left\{ -\frac{1}{2} \|\Sigma_t^{-1/2} (y - m_t(x))\|^2 \right\} g(y) dy.$$

216 Indeed, it holds that $h_\psi(x, t) = \mathcal{T}_{T-t} e^{\psi(x)}$. Then, introducing

$$q(y|x) = \frac{1}{(2\pi)^{d/2} \sqrt{\det(\Sigma_T)}} \exp \left\{ -\frac{1}{2} \|\Sigma_T^{-1/2} (y - m_T(x))\|^2 \right\},$$

217 we note that

$$\rho_T^\psi(y) = \int_{\mathbb{R}^d} \frac{q(y|x) e^{\psi(y)}}{\mathcal{T}_T e^{\psi(x)}} \rho_0(x) dx \quad (5)$$

218 is the marginal density of X_T^ψ , the endpoint of a random process X_t^ψ governed by h_ψ :

$$dX_t^\psi = b(m - X_t^\psi) dt + \nabla \log \left(\mathcal{T}_{T-t} e^{\psi(X_t^\psi)} \right) dt + \Sigma^{1/2} dW_t, \quad X_0^\psi \sim \rho_0.$$

219 If the Schrödinger potential ν_T admits a density e^{ψ^*} with respect to the Lebesgue measure, then the
 220 optimally controlled process X_t^* solves the SDE

$$dX_t^* = b(m - X_t^*) dt + \nabla \log \left(\mathcal{T}_{T-t} e^{\psi^*(X_t^*)} \right) dt + \Sigma^{1/2} dW_t, \quad X_0^* \sim \rho_0.$$

221 Finally, it is well known that the unique stationary (invariant) distribution of X_t^0 is Gaussian, that is,
 222 X_t^0 converges to X_∞^0 in distribution as $t \rightarrow \infty$ with $X_\infty^0 \sim \mathcal{N}(m, \Sigma/(2b))$. Since the parameters of
 223 the limiting distribution do not depend on the starting point, $\mathcal{T}_\infty g(x) \equiv \mathcal{T}_\infty g$ is a constant.

224 **Other notations** The notation $f \lesssim g$ or $g \gtrsim f$ means that $f = \mathcal{O}(g)$. Besides, we often replace
 225 $\max\{a, b\}$ and $\min\{a, b\}$ by shorter expressions $a \vee b$ and $a \wedge b$, respectively. For any $s \geq 1$, the
 226 Orlicz ψ_s -norm of a random variable ξ is defined as

$$\|\xi\|_{\psi_s} = \inf \left\{ u > 0 : \mathbb{E} e^{|\xi|^s/u^s} \leq 2 \right\}.$$

227 Finally, given $p \geq 1$ and a probability density ρ , the weighted L_p -norm of a function f is defined as
 228 $\|f\|_{L_p(\rho)} = (\mathbb{E}_{\xi \sim \rho} |f(\xi)|^p)^{1/p}$. Given two probability densities $\rho_0 \ll \rho_1$ on $\mathbb{R}^d$, the Kullback-Leibler
 229 divergence between them is defined as $\text{KL}(\rho_0, \rho_1) = \mathbb{E}_{\xi \sim \rho_0} \log(\rho_0(\xi)/\rho_1(\xi))$.

230 4 Main result

231 In the present section, we discuss statistical properties of the empirical risk minimizer $\hat{\psi}$ defined
 232 in (3). In particular, Theorem 1 provides a Bernstein-type upper bound on its excess KL-risk. We
 233 impose the following assumptions. First, as we announced before, we use the Ornstein-Uhlenbeck
 234 process X_t^0 as the reference one.

235 **Assumption 1.** *The base process X^0 solves the following SDE*

$$dX_t^0 = b(m - X_t^0) dt + \Sigma^{1/2} dW_t, \quad 0 \leq t \leq T.$$

236 where $b > 0$, $m \in \mathbb{R}^d$, Σ is a positive definite symmetric matrix of size $d \times d$, and W is a d -
 237 dimensional Brownian motion.

238 Main properties of the Ornstein-Uhlenbeck process were discussed in the previous section. Second,
 239 we suppose that the target density ρ_T^* meets the following requirements.

240 **Assumption 2.** *The target distribution at time T admits a bounded density ρ_T^* with respect to the*
 241 *Lebesgue measure such that*

$$\rho_T^*(x) \leq \rho_{\max} \quad \text{for all } x \in \mathbb{R}^d.$$

242 Moreover, the target distribution ρ_T^* is sub-Gaussian with variance proxy v^2 , that is,

$$\mathbb{E}_{Y \sim \rho_T^*} e^{u^\top Y} \leq e^{v^2 \|u\|^2/2} \quad \text{for any } u \in \mathbb{R}^d. \quad (6)$$

Assumption 2 is very mild. Despite the fact that we deal with logarithmic loss, we do not require ρ_T^* to be bounded away from zero. We do not even require its support to be compact. This significantly complicates the proof of the excess KL-bound and poses nontrivial technical challenges. Let us note that the condition 6 yields that $\mathbb{E}_{Y \sim \rho_T^*} Y = 0$. However, it does not diminish generality of our setup.

The remaining assumptions concern properties of the class of log-potentials Ψ . First, we assume that admissible log-potentials $\psi(x)$ are bounded from above and behave as $\mathcal{O}(\|x\|^2)$ as x tends to infinity.

Assumption 3. *There exist non-negative constants Λ and M such that*

$$-\Lambda \left\| \Sigma^{-1/2}(x - m) \right\|^2 - M \leq \psi(x) \leq M \quad \text{for all } x \in \mathbb{R}^d \text{ and } \psi \in \Psi.$$

Moreover, for any $\psi \in \Psi$, it holds that $\mathcal{T}_\infty \psi = \mathbb{E} \psi(X_\infty) = 0$.

The condition $\mathcal{T}_\infty \psi = 0$ appears because of the fact that the Schrödinger potentials ν_0 and ν_T (see (1)) are defined up to a multiplicative constant. The requirement $\mathcal{T}_\infty \psi = 0$ is nothing but a normalization. Second, we assume that Ψ is parametrized by a finite-dimensional parameter $\theta \in \mathbb{R}^D$:

$$\Psi = \{\psi_\theta : \theta \in \Theta\},$$

where Θ is a subset of a D -dimensional cube $[-R, R]^D$ and each function ψ_θ maps $\mathbb{R}^d$ onto $\mathbb{R}$. We suppose that the parametrization is sufficiently smooth in the following sense.

Assumption 4. *There exists $L \geq 0$ such that*

$$|\psi_\theta(x) - \psi_{\theta'}(x)| \leq L(1 + \|x\|^2) \|\theta - \theta'\|_\infty \quad \text{for all } \theta, \theta' \in \Theta \text{ and all } x \in \mathbb{R}^d.$$

Assumptions 3 and 4 are quite general. We provide two examples when they hold. First, in a recent paper [Korotin et al., 2024], the authors model $e^{\psi(x)}$ as a Gaussian mixture. Let $\alpha_1, \dots, \alpha_K$ be non-negative numbers such that $\alpha_1 + \dots + \alpha_K = 1$ and consider

$$e^{\psi(x)} = e^{-C} \sum_{k=1}^K \alpha_k \varphi_{m_k, \Sigma_k}(x), \quad \text{where} \quad \varphi_{m_k, \Sigma_k}(x) = \frac{e^{-\|\Sigma_k^{-1/2}(x - m_k)\|^2/2}}{(2\pi)^{d/2} \det(\Sigma_k)^{1/2}}.$$

Here C is a normalizing constant which ensures that $\mathcal{T}_\infty \psi = 0$. In this situation, the parameter θ consists of all α_k 's and all components of m_k 's and Σ_k 's, $k \in \{1, \dots, K\}$. If the smallest eigenvalues of $\Sigma_1, \dots, \Sigma_K$ are bounded away from zero uniformly over $k \in \{1, \dots, K\}$, then $e^{\psi(x)}$ is bounded. On the other hand, if K is fixed, there is a component with a weight at least $1/K$. Without loss of generality, we assume that it is the first one. Then

$$\psi(x) \geq -C + \log(\alpha_1 \varphi_{m_1, \Sigma_1}(x)) \geq -C - \log K - \frac{1}{2} \left\| \Sigma_1^{-1/2}(x - m_1) \right\|^2,$$

and we conclude that Assumption 3 is satisfied. Verification of the Assumption 4 is straightforward once we assume that the weight of each component is bounded away from zero, and the norms $\|m_k\|$, $\|\Sigma_k\|$, and $\|\Sigma_k^{-1}\|$ are bounded uniformly over $k \in \{1, \dots, K\}$ (which is the case in [Korotin et al., 2024]). Second, Assumptions 3 and 4 will be fulfilled if one deals, for example, with a class of truncated feedforward neural networks with bounded weights and ReLU activations. It is known that (see [Schmidt-Hieber, 2020, Lemma 5]) they are Lipschitz with respect to each weight, and the Lipschitz constant grows linearly with $\|x\|$. More generally, [Conforti, 2024] analyzed semiconvexity properties of the Schrödinger potentials under rather mild assumptions on the marginals.

We are ready to formulate the main result of this section.

Theorem 1. *Let ρ_0 be the density of the standard Gaussian distribution $\mathcal{N}(0, I_d)$. Grant Assumptions 1, 2, 3, and 4. Assume that T is sufficiently large in a sense that*

$$bT \geq (5 + \log d) \vee \log(160b(\mathfrak{v}^2 \vee 1) \|\Sigma^{-1}\|).$$

Let $\hat{\psi}$ be defined in (3) and let $\hat{\rho}_T$ be the corresponding density of $X_T^{\hat{\psi}}$. Then, for any $\delta \in (0, 1/2)$, with probability at least $1 - 2\delta$, it holds that

$$\text{KL}(\rho_T^*, \hat{\rho}_T) - \inf_{\psi \in \Psi} \text{KL}(\rho_T^*, \rho_T^\psi) \lesssim \sqrt{\Upsilon(n, \delta) \inf_{\psi \in \Psi} \text{KL}(\rho_T^*, \rho_T^\psi)} + \Upsilon(n, \delta),$$

where

$$\Upsilon(n, \delta) = (\Lambda d + M + d) \left(d + \log \frac{RLn}{\delta} + (M \vee \log \Lambda) \sqrt{d} e^{-bT} \right) \frac{D \log n}{n}.$$

The hidden constant behind $\lesssim$ depends on Σ , m , b , and $\mathfrak{v}$ only.

280 In Theorem 1, we assume that ρ_0 is the density of $\mathcal{N}(0, I_d)$. Though it is a standard choice of
 281 initial distribution in practice, we would like to emphasize that unbounded support of ρ_0 significantly
 282 complicates the proof and makes the problem even more challenging.

283 The problem of Schrödinger potential estimation was also studied in [Korotin et al., 2024] and
 284 [Pooladian and Niles-Weed, 2024]. In [Korotin et al., 2024], the authors suggest an algorithm called
 285 Light Schrödinger Bridge, which is based on minimization of the empirical KL-divergence between
 286 entropic optimal transport plans. This slightly differs from our setup, since we aim to minimize
 287 empirical KL-divergence between marginal endpoint distributions. The reason is that [Korotin,
 288 Gushchin, and Burnaev] [2024] are motivated by the style transfer task, where the initial distribution
 289 is also unknown. In contrast, we focus on generative modelling where the initial distribution ρ_0
 290 is available to learner. In [Korotin et al., 2024] Theorem A.1], the authors consider the case when
 291 admissible potentials are Gaussian mixtures with K components. Assuming that both initial and
 292 finite distributions have a compact support, they prove a $\mathcal{O}(n^{-1/2})$ upper bound on the Rademacher
 293 complexity of such class. On the other hand, we allow the support of ρ_0 and ρ_T^* to be unbounded.
 294 Besides, the rate of convergence presented in Theorem 1 may be much faster than $\mathcal{O}(n^{-1/2})$ if the
 295 target distribution is close to $\{\rho_T^\psi : \psi \in \Psi\}$. In the realizable case (that is, $\rho_T^* \in \{\rho_T^\psi : \psi \in \Psi\}$) the
 296 right-hand side in Theorem 1 becomes $\mathcal{O}(\log^2 n/n)$. Finally Theorem 1 provides a high-probability
 297 upper bound on the excess risk while the result of [Korotin et al., 2024] holds in expectation. In
 298 [Pooladian and Niles-Weed, 2024] the authors study properties of a plug-in Sinkhorn-based estimator.
 299 Similarly to [Korotin et al., 2024], they consider the case of compactly supported initial and target
 300 measures. However, they assume that these measures are supported on smooth k -dimensional
 301 submanifolds. They derive a $\mathcal{O}(n^{-1/2} + (T - \tau)^{-k-2}n^{-1})$ bound on the squared total variation
 302 distance between *path measures* up to moment $\tau < T$. Unfortunately, the second term grows very
 303 fast when τ approaches T , and there are no guarantees whether the marginal endpoint distributions
 304 will be close to each other.

305 In Theorem 1, we focus on the statistical error leaving study of the approximation out of the scope
 306 of the present paper. The reason is that there are few results on properties of the true log-potential
 307 $\psi^*(x) = \log(\nu_T(dx)/dx)$. However, we would like to note that, according to our findings (see
 308 Lemma B.2 and (5)), if ψ^* fulfils Assumption 3 then for any $\psi \in \Psi$ and $y \in \mathbb{R}^d$

$$\begin{aligned} \log \frac{\rho_T^*(y)}{\rho_T^\psi(y)} &\lesssim |\psi(y) - \psi^*(y)| \\ &\quad + (\mathcal{T}_\infty |\psi - \psi^*|)^{1/\mathcal{K}(T)} \|\Sigma^{-1/2}(y - m)\|^{2-2/\mathcal{K}(T)} e^{\mathcal{O}(e^{-bT} \|\Sigma^{-1/2}(y-m)\|^2)}, \end{aligned}$$

309 where $1 \leq \mathcal{K}(T) \leq 1 + \mathcal{O}(\sqrt{de^{-bT}})$. In the proof of Theorem 1 (see Step 4), we show that the
 310 expectation

$$\mathbb{E}_{Y \sim \rho_T^*} \left\| \Sigma^{-1/2}(Y - m) \right\|^{2-2/\mathcal{K}(T)} e^{\mathcal{O}(e^{-bT} \|\Sigma^{-1/2}(Y-m)\|^2)}$$

311 is finite, provided that $bT \geq (5 + \log d) \vee \log(160b(\vee^2 \vee 1) \|\Sigma^{-1}\|)$. This allows us to relate the
 312 KL-divergence between ρ_T^* and ρ_T^ψ with the distances between the corresponding log-potentials:

$$\text{KL}(\rho_T^*, \rho_T^\psi) \lesssim \|\psi - \psi^*\|_{L_1(\rho_T^*)} + (\mathcal{T}_\infty |\psi - \psi^*|)^{1/\mathcal{K}(T)}.$$

313 5 Proof sketch of Theorem 1

314 In this section, we discuss main ideas used in the proof of Theorem 1. Rigorous derivations are
 315 deferred to Appendix A. Since the proof is quite long, we split it into several steps.

316 **Step 1: log-density properties.** Let us note that Assumptions 3 and 4 concern properties of
 317 log-potentials $\psi \in \Psi$ while empirical risks include marginal densities ρ_T^ψ . For this reason, before we
 318 consider the empirical process

$$\frac{1}{n} \sum_{i=1}^n \log \frac{\rho_T^*(Y_i)}{\rho_T^\psi(Y_i)} - \text{KL}(\rho_T^*, \rho_T^\psi), \quad \psi \in \Psi,$$

we have to study the random variables $\log(\rho_T^*(Y_i)/\rho_T^\psi(Y_i))$, $1 \leq i \leq n$. Using basic properties of the Ornstein-Uhlenbeck operator, we show that

$$-\log \rho_T^\psi(y) \lesssim -\psi(y) + \|\Sigma^{-1/2}(y - m)\|^2.$$

In view of Assumption 3, this means that $-\log \rho_T^\psi(y)$ grows as fast as a quadratic function. Since the target distribution is sub-Gaussian and has a bounded density, this yields that the random variables $\log(\rho_T^*(Y_i)/\rho_T^\psi(Y_i))$, $1 \leq i \leq n$, are sub-exponential. More specifically, applying Lemma C.3 we obtain the following upper bound on their Orlicz norm:

$$\left\| \log \frac{\rho_T^*(Y_i)}{\rho_T^\psi(Y_i)} \right\|_{\psi_1} \lesssim \Lambda d + M + d \quad \text{for all } i \in \{1, \dots, n\}.$$

Step 2: ε -net argument and Bernstein's inequality. The result obtained on the first step allows us to use concentration inequalities for sub-exponential random variables. Let us fix $\varepsilon \in (0, R)$ and let Θ_ε stand for the minimal ε -net of Θ with respect to the ℓ_∞ -norm. We denote the set of corresponding log-potentials by Ψ_ε :

$$\Psi_\varepsilon = \{\psi_\theta : \theta \in \Theta_\varepsilon\}.$$

Using Bernstein's inequality for unbounded random variables (see, for instance, [Lecué and Mitchell, 2012], Proposition 5.2) and the union bound, we obtain that

$$\left| \text{KL}(\rho_T^*, \rho_T^\psi) - \frac{1}{n} \sum_{i=1}^n \log \frac{\rho_T^*(Y_i)}{\rho_T^\psi(Y_i)} \right| \lesssim \sqrt{\text{Var} \left(\log \frac{\rho_T^*(Y_1)}{\rho_T^\psi(Y_1)} \right) \frac{\log(2|\Psi_\varepsilon|/\delta)}{n}} + \frac{(\Lambda d + M + d) \log n \log(2|\Psi_\varepsilon|/\delta)}{n}$$

with probability at least $(1 - \delta)$ simultaneously for all $\psi \in \Psi_\varepsilon$.

Step 3: bounding the loss variance. One of the key ingredients in the proof of Theorem 1, which allows us to hope for faster rates of convergence than $\mathcal{O}(n^{-1/2})$, is analysis of the variance of $\log(\rho_T^*(Y_1)/\rho_T^\psi(Y_1))$, $\psi \in \Psi$. Despite the fact that the admissible log-potentials may be unbounded, we are still able to show that the class Ψ satisfies a Bernstein-type condition

$$\text{Var} \left(\log \frac{\rho_T^*(Y_1)}{\rho_T^\psi(Y_1)} \right) \lesssim (\Lambda d + M + d) \log n \left(\text{KL}(\rho_T^*, \rho_T^\psi) + \frac{1}{n} \right).$$

Steps 4 and 5: from ε -net to a uniform Bernstein-type bound. The hardest and technically involved part of the proof is to show that the losses $\log(\rho_T^*(y)/\rho_T^\psi(y))$ and $\log(\rho_T^*(y)/\rho_T^\phi(y))$ do not differ too much, once the corresponding log-potentials ψ and ϕ are close to each other. This follows from Lemma B.2 which relies on properties of the Ornstein-Uhlenbeck operator established in and Lemma B.3. We would like to note that the unbounded support of the initial density ρ_0 significantly complicates the proof of Lemma B.2. Nevertheless, we prove that

$$\log \frac{\rho_T^\psi(y)}{\rho_T^\phi(y)} \lesssim |\psi(y) - \phi(y)| + (\mathcal{T}_\infty |\psi - \phi|)^{1/\mathcal{K}(T)} \|\Sigma^{-1/2}(y - m)\|^{2-2/\mathcal{K}(T)} e^{\mathcal{O}(e^{-bT} \|\Sigma^{-1/2}(y - m)\|^2)},$$

where $1 \leq \mathcal{K}(T) \leq 1 + \mathcal{O}(\sqrt{d}e^{-bT})$. Though the right-hand side depends exponentially on the squared norm of $\Sigma^{-1/2}(y - m)$, the coefficient $\mathcal{O}(e^{-bT})$ is quite small, which is enough for our purposes.

Steps 6 and 7: choice of ε and the final bound. The rest of the proof is quite standard. On Step 6, we choose an appropriate ε and obtain a uniform Bernstein-type inequality

$$\text{KL}(\rho_T^*, \rho_T^\psi) - \frac{1}{n} \sum_{i=1}^n \log \frac{\rho_T^*(Y_i)}{\rho_T^\psi(Y_i)} \lesssim \sqrt{\Upsilon(n, \delta) \text{KL}(\rho_T^*, \rho_T^\psi)} + \Upsilon(n, \delta),$$

where

$$\Upsilon(n, \delta) = (\Lambda d + M + d) \left(d + \log \frac{RLn}{\delta} + (M \vee \log \Lambda) \sqrt{d} e^{-bT} \right) \frac{D \log n}{n},$$

which holds simultaneously for all $\psi \in \Psi$ with probability at least $(1 - 2\delta)$. After that, we transform it into the desired excess risk bound and finish the proof.

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